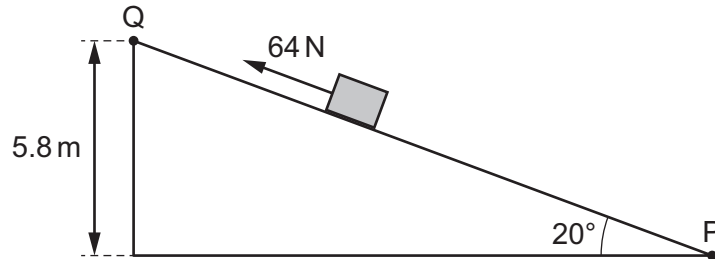


- 19** A box of mass 4.9 kg is pushed at a constant velocity from point P at the bottom of an inclined plane to point Q at the top.

The box is pushed by a force of 64 N acting parallel to the slope.

The slope is inclined at an angle of 20° to the horizontal and the box moves through a vertical height of 5.8 m.



What is the work done against the frictional force acting on the block between P and Q?

- A** 280 J **B** 810 J **C** 960 J **D** 1100 J

- 20** Which expression gives the efficiency of a system?

- A** $\frac{\text{total energy input}}{\text{useful energy output}}$
- B** $\frac{\text{useful energy output} + \text{wasted energy output}}{\text{total energy input}}$
- C** $\frac{\text{useful energy output}}{\text{total energy input}}$
- D** $\frac{\text{wasted energy output}}{\text{total energy input}}$

- 21** A student has a copper wire and a steel wire with equal lengths and cross-sectional areas.

The student hangs identical loads on the two wires.

The extensions of the two wires are different.

The student calculates the stress, strain and Young modulus of each wire.

Which row identifies with a tick (✓) the calculated values that are equal for both wires?

	stress	strain	Young modulus
A	✓		
B		✓	
C		✓	✓
D	✓		✓